

VLSI Technology and Design

Cadence Design Project

To: Prof. Jan Duotrelloigne

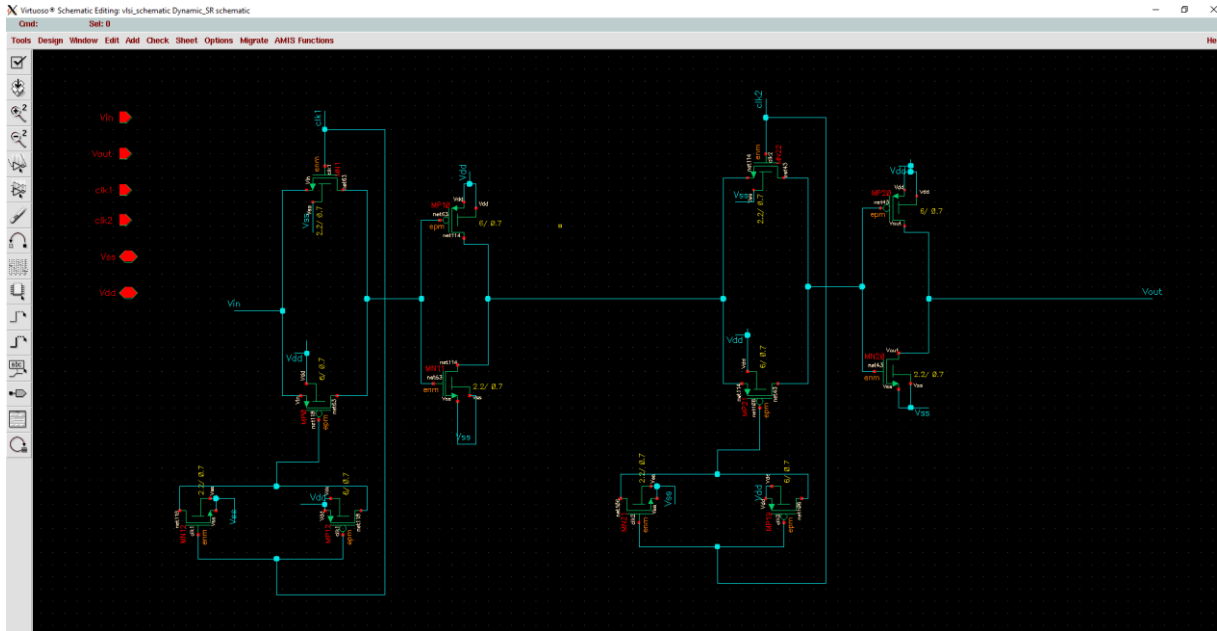
Done by: Omar Moussa

Report content

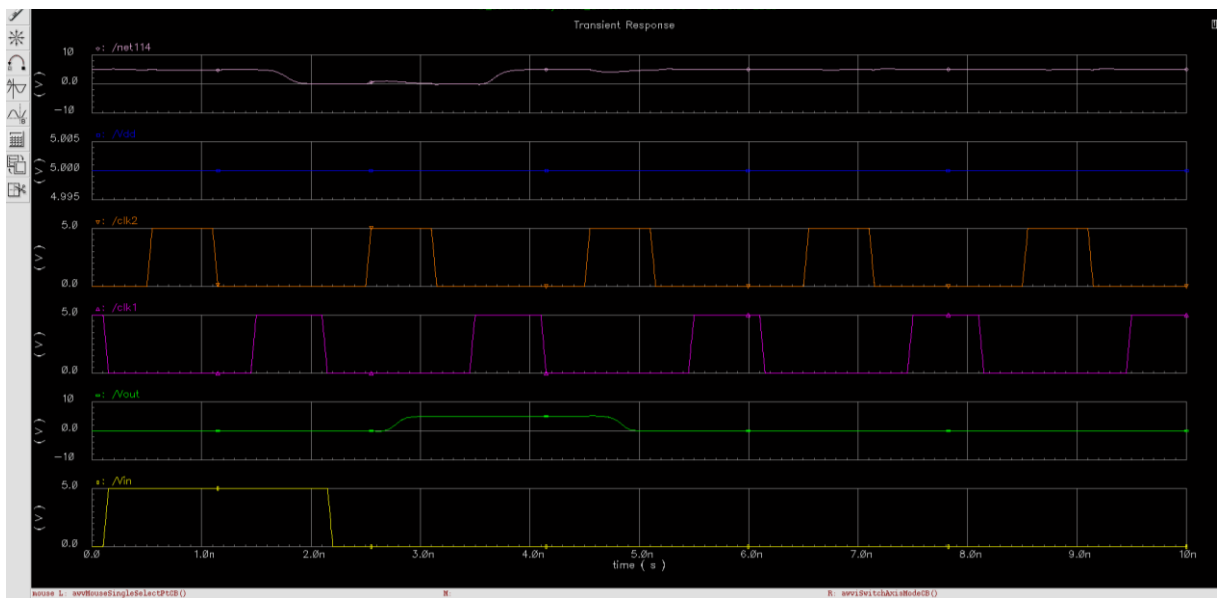
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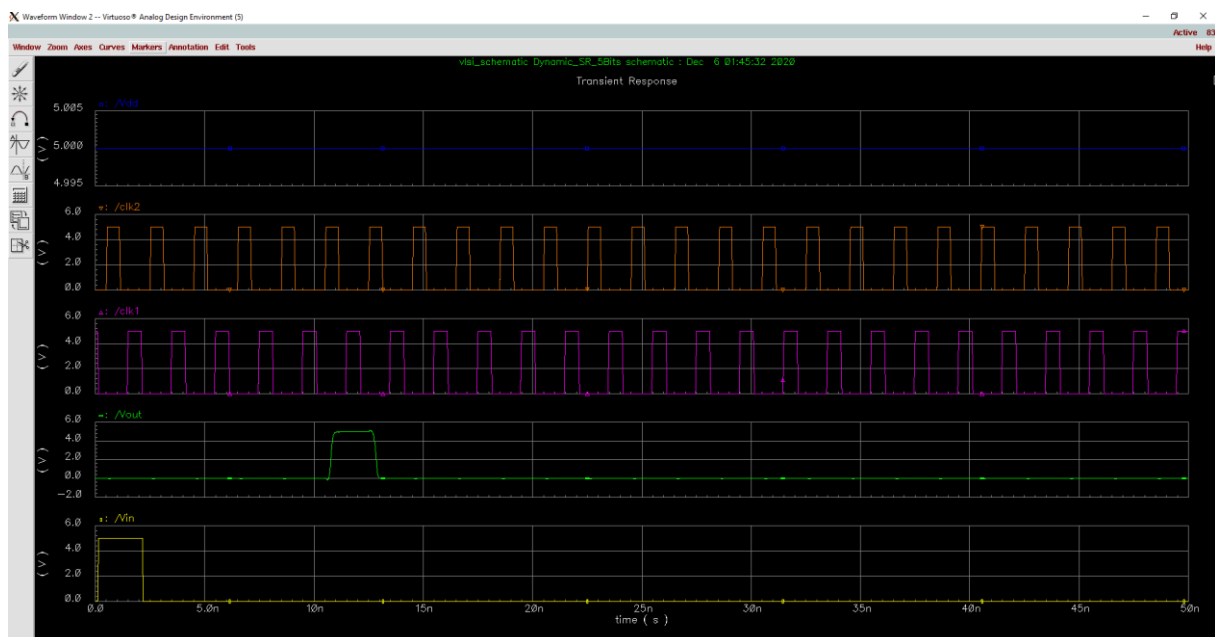
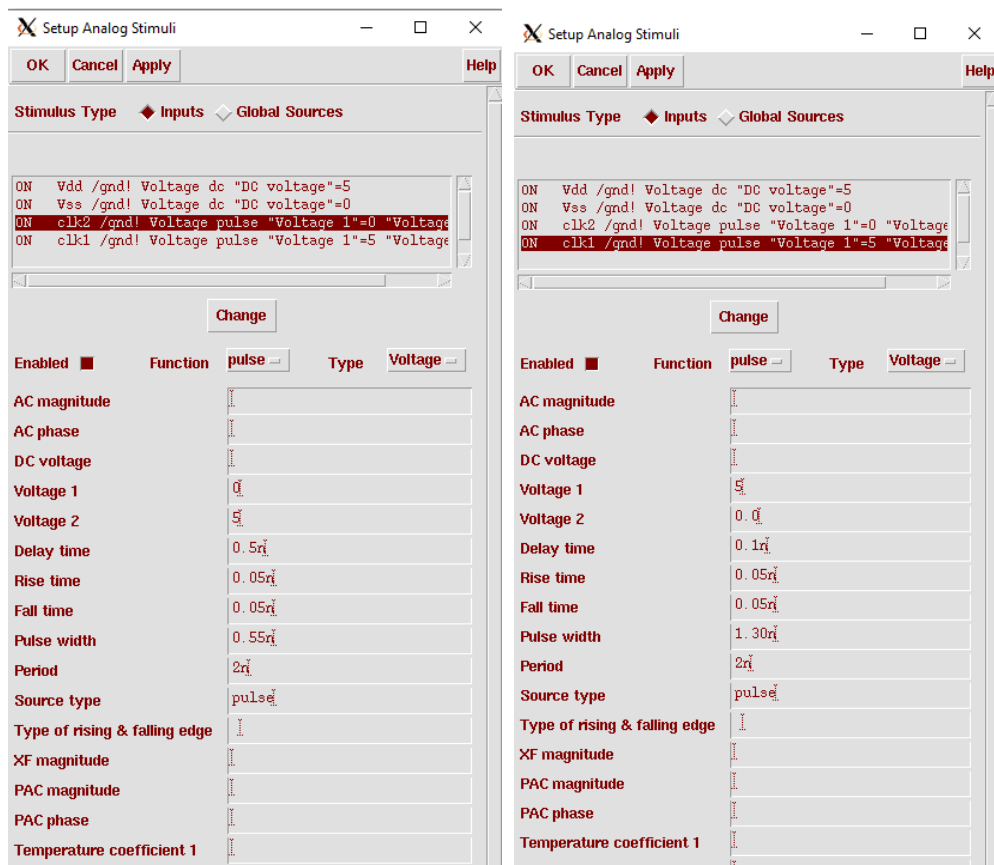
Cadence Design Project

1- Schematic design: Question A:

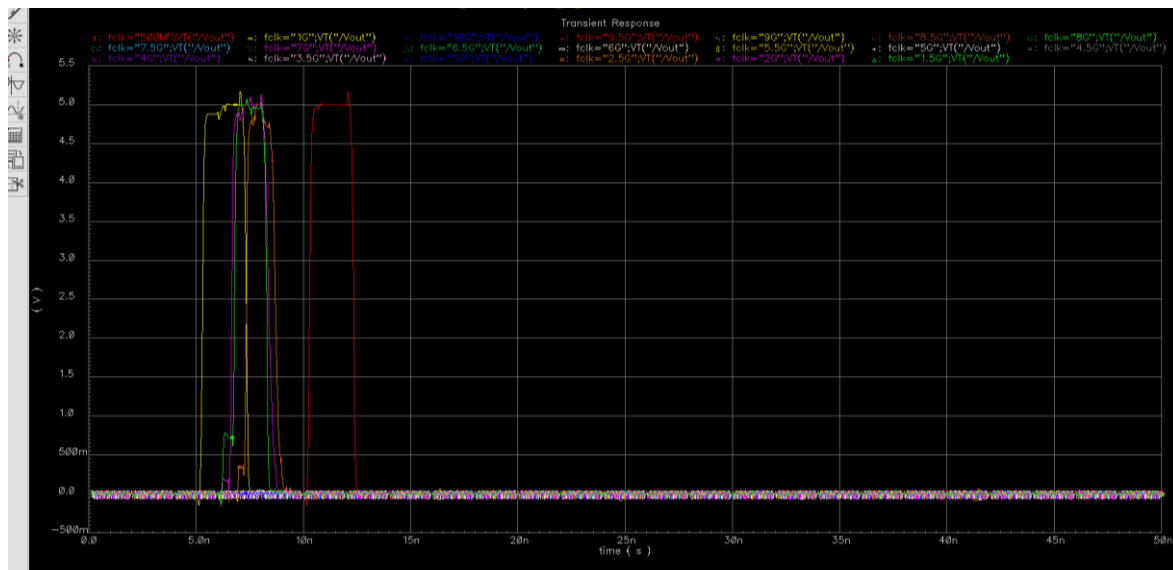


- Spectrum simulation for 1 bit for elementary shift register

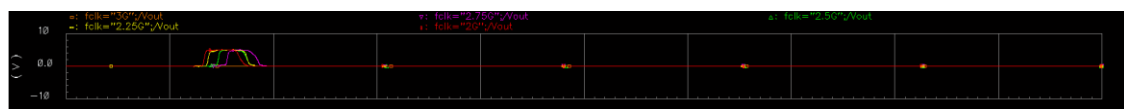




Parametric sweep from 500M to 10G

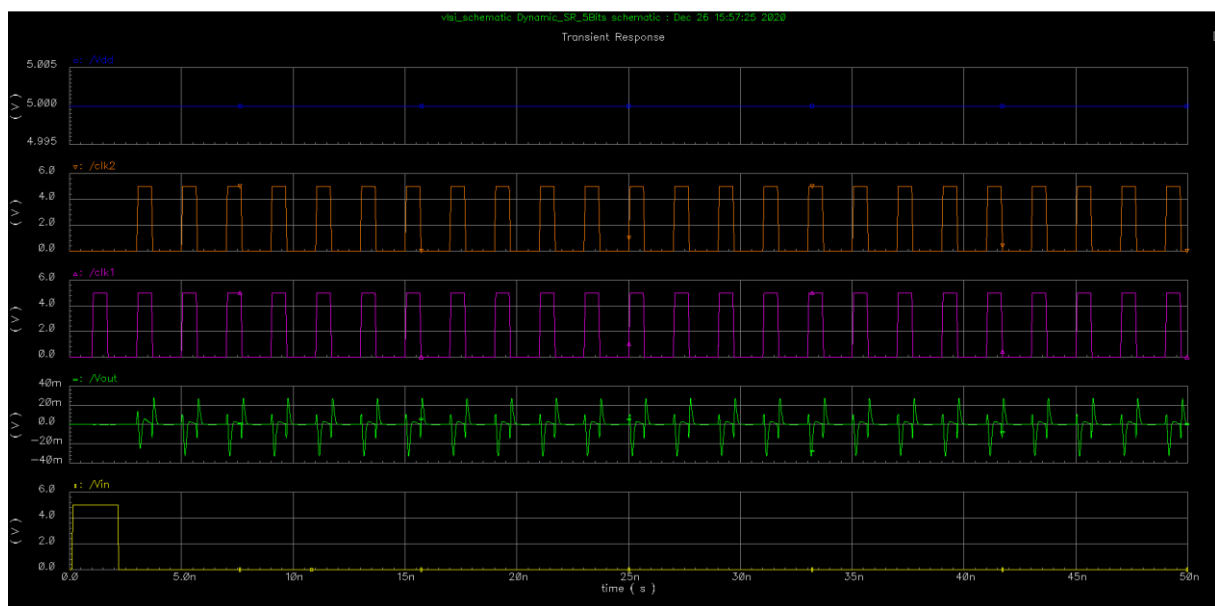


- Then a Parametric sweep between 2.5 – 3G



- We got 2.75 G is the max operating freq, the min freq is 500 MHz.
- $600 \times 800 \times 50 = 24 \text{ M Hz}$ which is not sufficient to be operated as the SR's min operating freq is more than the 24 M Hz.

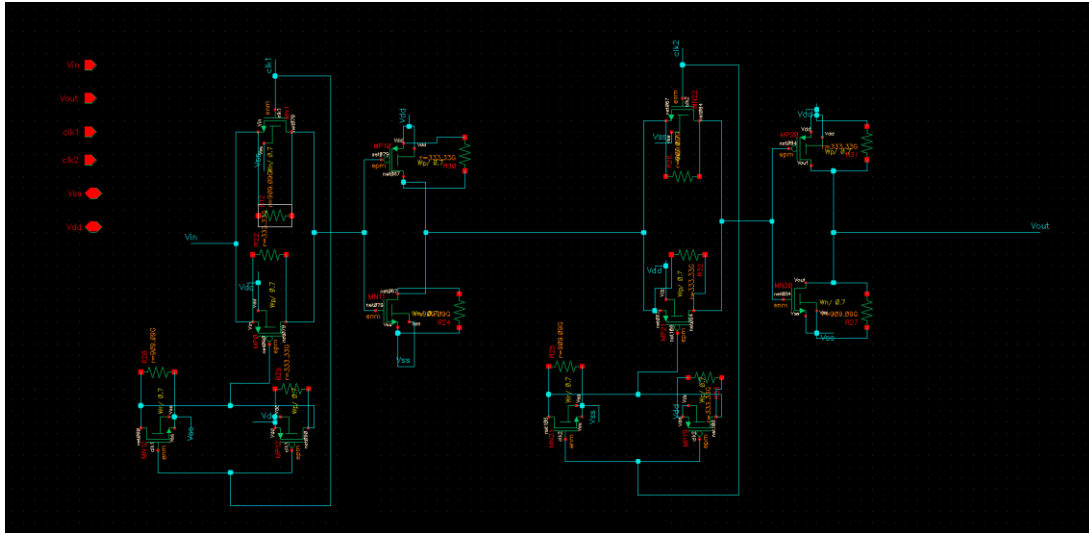
By overlapping clocks:



There will be no output because there's no bit shifting happening due to the clock overlapping, physically that switches are switched ON, OFF sharing the same time, which didn't give the time to transfer the register to the next stage.

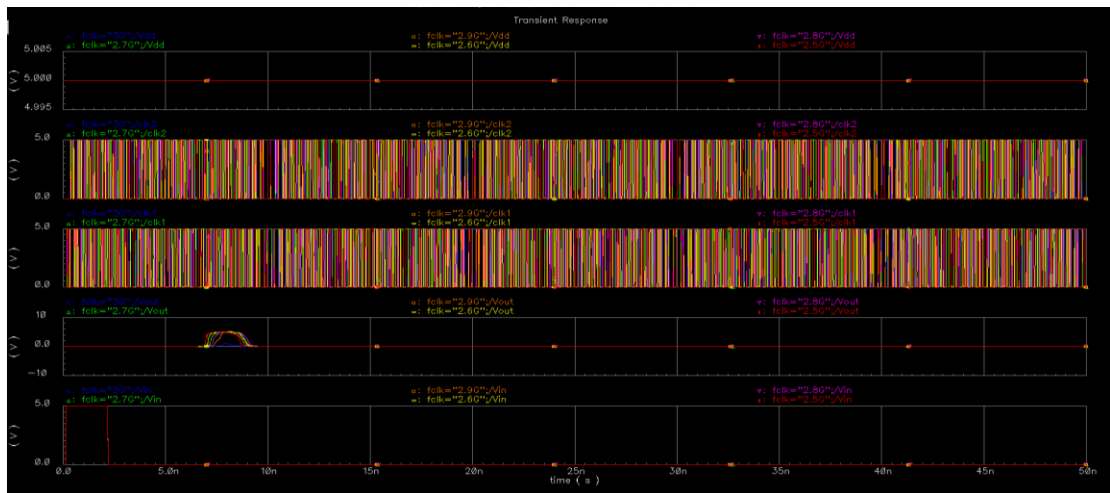
Question B:

- Adding illumination resistances of $R_n = 909.09 \text{ Kohms}$, $R_p = 333.33 \text{ Kohms}$

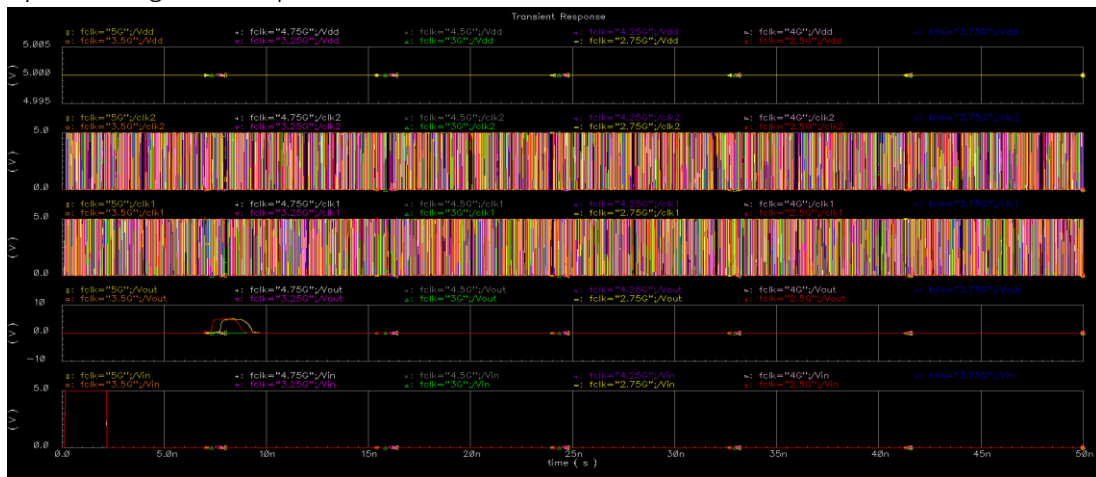


Parametric sweep:

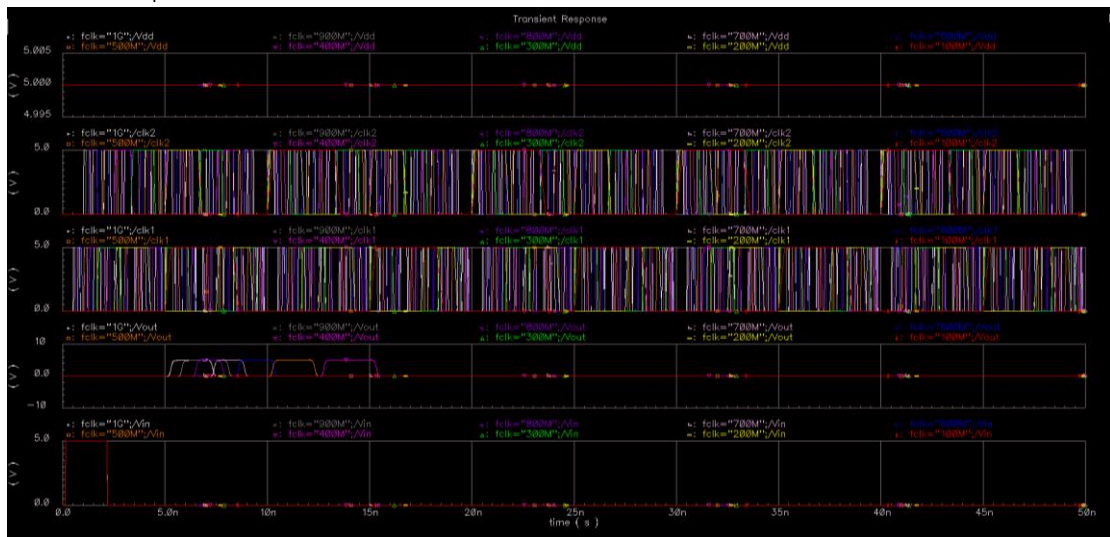
- The maximum frequency under the illuminated conditions is 2.7 G



- By narrowing the sweeps:



- The min freq is 400MHz

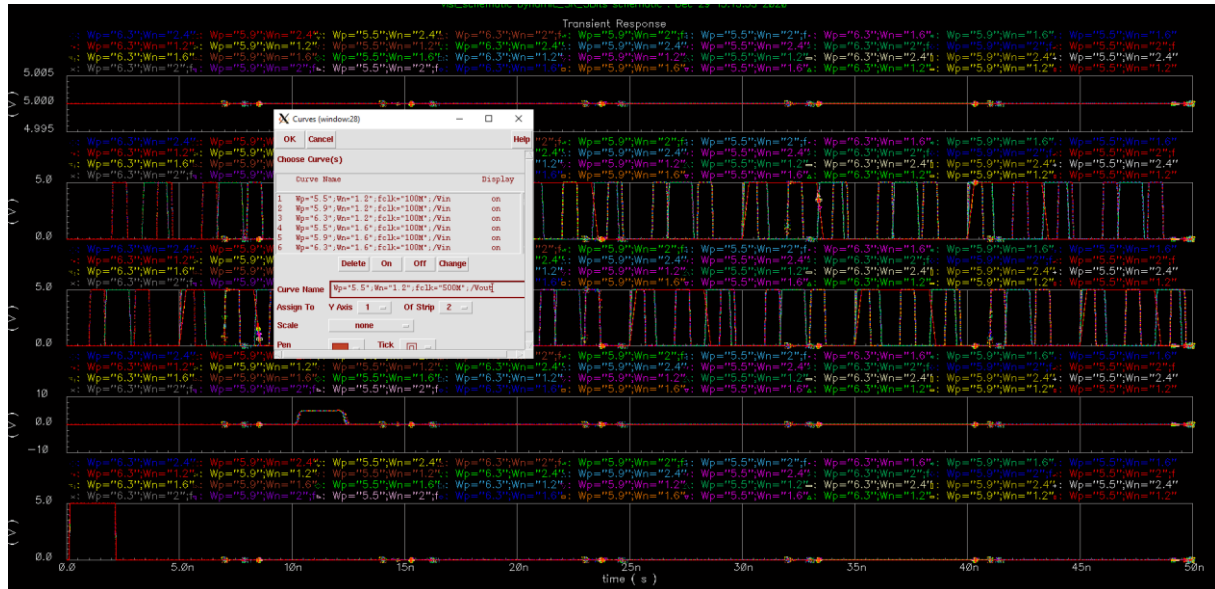


- Difference compared to the situation without any light falling onto the chip, is the output's amplitude got dropped a bit probably that's because of the effect of the added resistance.
- Column driver of the 600x800 SVGA LCOS micro-display with 50Hz refresh rate shouldn't operate properly with the SR.

Question C:

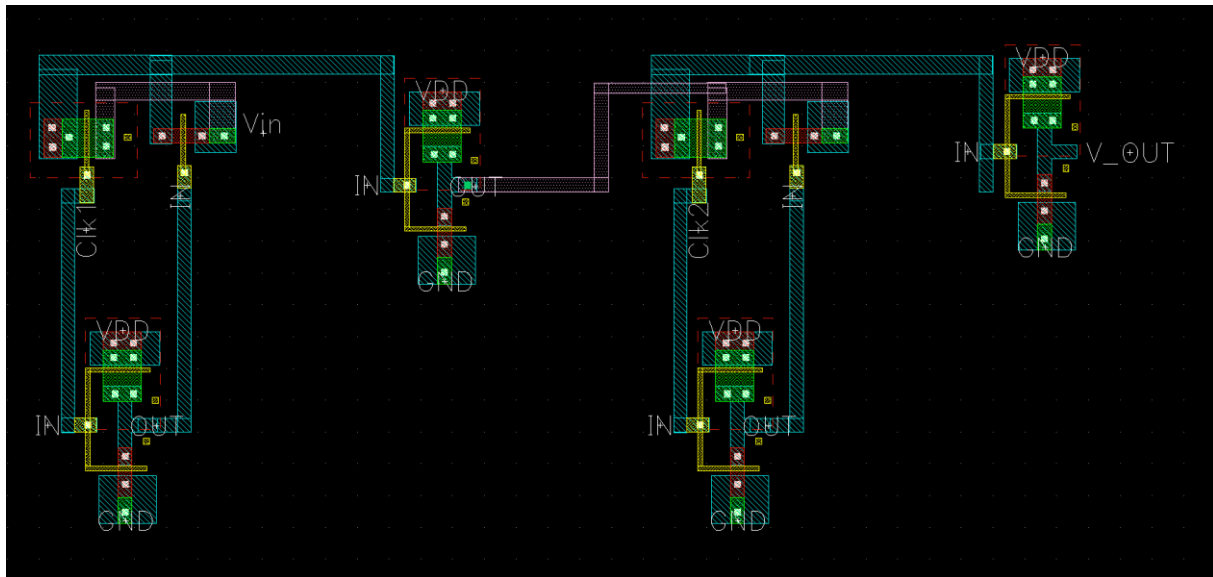
By running parametric sweep of variables widths W_n , W_p @ 500MHz ($\sim 3 \cdot f_{clk}$), we got dimensions using parametric sweep: $W_p=5.5\mu m$ and $W_n=1.2\mu m$.

Which is logic because it increase the resistance value of a transistor which is in parallel with an added illuminated resistance resistance, then the charging time increase which result in to decrease the max operating frequency to be @ 500MHz.

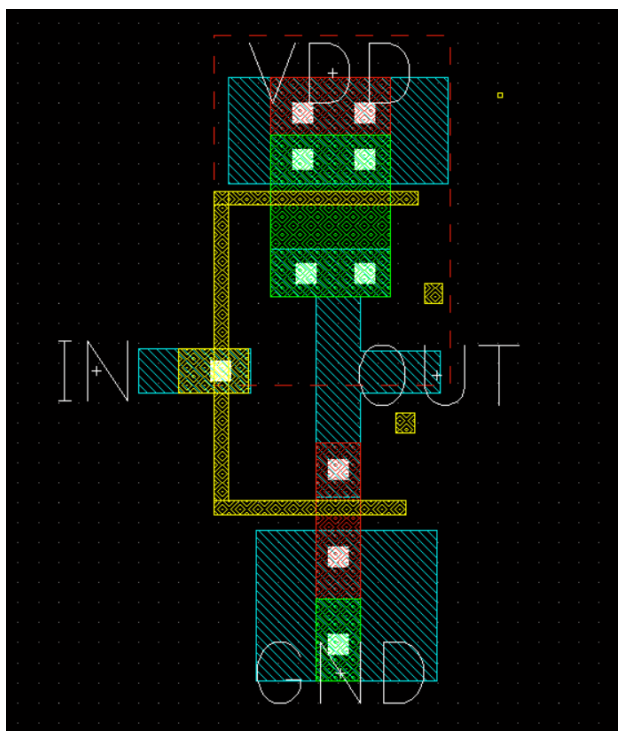


Schematic Layout

All layers:



Layout of One inverter



Layout of One transmission gate:

